

Xinglang Zhang

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Education

Huazhong University of Science and Technology (HUST)

Sept 2024 – Jun 2028

B.S. in Cyber Science and Engineering

- **GPA:** 4.80/5.0 **Weighted Avg:** 94.51 **Rank:** 1/108
- **Selected Coursework:** Calculus A (I/II) (A+), Linear Algebra (A+), Discrete Math (A+), Data Structures (A+), Assembly Language (A+), Probability Theory (A+), Digital Circuit (A+), Database Systems (A+).

Research Agenda

Reliable LLM Reasoning. When and why does reasoning break down, and how can structured or neuro-symbolic mechanisms make it more reliable?

Test-Time Learning & Self-Evolving Agents. How can deployed models learn from their own experience without gold supervision, and decide what, when, and how to learn as their state evolves?

Human-AI Preference & Social Intelligence. Where does model behavior diverge from human preferences and social signals, and how can these gaps inform model design?

Research Experience

Adaptive Learning & Self-Evolving Agents

TRACE: Transfer-Aware Test-Time Training for Tool-Using LLMs

Jul 2026 – Present

Research Intern, BLENDER Lab, University of Illinois Urbana-Champaign (UIUC)

Mentor: [Cheng Qian](#)

- **Question.** Test-time training can learn from self-generated tool-use practices, but a limited update budget makes experience selection essential. Formulated the question: *which experience is worth learning now, given what the model has already learned?*
- **Finding.** Characterized *Transfer Value* as the benefit of learning from an experience for a target distribution. Established that this utility is **relational and state-dependent**: the same experience can help one query and harm another, and its value can reverse as the solver updates. Demonstrated cross-query transfer and non-monotonic changes in utility. In a controlled probe, **30 of 32** initially beneficial experiences became harmful after 32 updates; some became useful again later, motivating repeated selection rather than a fixed ranking.
- **Method.** Developed *TRACE* to combine *Target Relevance*, *Current-State Transferability*, and *Marginal Complementarity*. Temporary updates and gold-free probes allocate the learning budget; complementary practices are selected, learned through persistent LoRA-SFT, and re-evaluated as the solver evolves.
- **Evidence.** Under matched permanent update budgets, probing the updated solver reached **38.1 Exact** versus **30.4** with stale-state probes. Reordering the same selected experiences degraded performance, showing that both *what* and *when* to learn matter. Across four tool-use benchmarks and four main models, improved Overall Exact accuracy by **13.10–24.42 percentage points** over frozen models and consistently outperformed static selection. These findings frame effective TTT as **dynamic curriculum formation**. First-author ACL 2027 submission, under review [3].

Reliable LLM Reasoning

Logical Phase Transitions: Diagnosing and Mitigating Reasoning Collapse

Aug 2025 – Jan 2026

Project Leader, Intelligent Media Computing and Network Security Lab, HUST

Advisor: [Zikai Song](#)

- Investigated where reasoning becomes unreliable as *logical complexity* increases, asking whether collapse follows identifiable regimes that aggregate benchmark scores obscure.
- **LoCM and NSA-LR.** Introduced the *Logical Complexity Metric (LoCM)* and *NSA-LR*, pairing natural-language statements with explicit first-order-logic representations and reasoning chains. Enabled complexity analysis beyond coarse hop-count estimates.
- Identified *Logical Phase Transitions*: accuracy remains relatively stable within complexity regimes, then drops sharply across critical intervals. Observed this pattern across open- and closed-source LLMs.
- Developed *Neuro-Symbolic Curriculum Tuning*, combining natural-language/symbolic alignment with complexity-aware training around the observed transition regions to progressively strengthen reasoning at higher complexity.
- Across five benchmarks, obtained average accuracy gains of **+1.26** with naive prompting and **+3.95** with CoT, alongside improved generalization to unseen logical compositions. Connected failure diagnosis to curriculum design. Co-first-author paper, **ACL 2026 Main** [1].

Research Experience (continued)

Reliable LLM Reasoning (continued)

From LogicAgent to HexLogicAgent: Structuring Meaning Before Deduction

Research Assistant, Intelligent Media Computing and Network Security Lab, HUST

Advisor: Zikai Song

LogicAgent: Feb – Aug 2025 HexLogicAgent: Mar – Jul 2026

- Investigated how *semantic complexity* interacts with *logical depth*: valid deduction depends on first interpreting abstract, ambiguous, or conflicting propositions correctly.
- **LogicAgent**. Contributed to a semiotic-square-guided framework that structures alternative meanings before automated deduction and reflective verification. Introduced *RepublicQA* to jointly evaluate semantic and logical difficulty; achieved **+6.25%** average improvement on RepublicQA and **+7.05%** across four mainstream benchmarks. Co-authored **ACL 2026 Main / Oral** paper [2].
- The square leaves mixed and partially compatible meanings underrepresented. Identified incomplete semantic representations as a major source of failures despite apparently valid deduction.
- **HexLogicAgent**. Extended the semantic structure from a square to a logical hexagon, explicitly representing uniform and mixed cases before deduction and verification. Developed a formal account of semantic opposition and completeness under stated first-order-logic assumptions.
- HexLogicAgent improved reliability across five benchmarks and delayed degradation as logical complexity increased. Established **semantic organization before inference** as a focus of this research line. Co-first-author *Artificial Intelligence* submission [5].

Human-AI Preference & Social Intelligence

OMRA: Diagnosing and Reducing Synthetic-Real Preference Misalignment

Jan – May 2026

Project Leader, Intelligent Media Computing and Network Security Lab, HUST

Advisor: Zikai Song

- Asked whether the qualities LLMs associate with high engagement match real user responses. Studied **1.17 million answers to 25,978 questions** on Zhihu, Quora, and Reddit; introduced *Ontological Preference Measurement* along logic, affect, and expression dimensions.
- Identified *logic overbinding*: as target engagement increases, LLMs add explicit reasoning structure, while observed user engagement is more strongly associated with affective and expressive salience.
- Developed *Ontology-Masked Reasoning Autoencoding (OMRA)* to mask and reconstruct over-explained spans while preserving stance, factual content, and coherence.
- Reduced marginal, geometric, and transfer gaps by **54.4% on average** across four LLM families. The results motivate grounding preference alignment in observed social signals. First-author NAACL 2027 submission [4]; all three reviewers gave **OA 3.5/5**, with a reported top-8% ranking.

Related collaborative work.

- Co-authored *IntervenSim* [7], modeling the co-evolution of interventions and crowd opinion; co-authored the *Social Intelligence Modeling* survey [6], connecting social perception with social simulation.

Publications

* Equal contribution.

[1] **Logical Phase Transitions: Understanding Collapse in LLM Logical Reasoning**

Xinglang Zhang*, Yunyao Zhang*, et al. | **ACL 2026 (Main; co-first author)** | [ArXiv] | [Code]

[2] **Semantic-Aware Logical Reasoning via a Semiotic Framework**

Yunyao Zhang, Xinglang Zhang, et al. | **ACL 2026 (Main; Oral)** | [ArXiv] | [Code]

[3] **TRACE: Transfer-Aware Test-Time Training for Tool-Using LLMs**

Xinglang Zhang, Cheng Qian | [Paper] | **ACL 2027, under review**

[4] **Too Good to Be Real? Diagnosing and Reducing the Gap Between AI Preference and Real User Engagement**

Xinglang Zhang, Yuanmeng Xiang, et al. | [ArXiv] | **NAACL 2027 submission**

[5] **Semiotic Logical Hexagon Theory for LLM Logical Reasoning**

Yunyao Zhang*, Xinglang Zhang*, et al. | [ArXiv] | [Code] | **Co-first author; Artificial Intelligence submission**

[6] **Social Intelligence Modeling: A Comprehensive Survey from Social Perception to Social Simulation**

Zikai Song, Xiajie Li, Yunyao Zhang, Xinglang Zhang, et al. | [Paper] | [GitHub]

[7] **IntervenSim: Intervention-Aware Social Network Simulation for Opinion Dynamics**

Yunyao Zhang, Zuocheng Ying, Xinglang Zhang | [ArXiv] | [Code]

[8] **EarthVerse: Benchmarking Scientific Agents Across Dynamic Earth Systems and Natural Hazards**

Zhiqing Cui, Xinxiang Yin, Yihong Tang, Xinglang Zhang, et al. | [Project Page] | [Paper]

Selected Funded Projects

- **HUST Undergraduate Natural Science Fund, 2026**
Trustworthy Enhancement for Complex Reasoning
Project Lead | CNY 50,000 (approx. US\$7,450)
- **College Student Innovation and Entrepreneurship Program (University Level)**
LLM Logical Reasoning through Philosophical Paradigms
Project Lead | CNY 4,000 (approx. US\$600)
- **College Student Innovation and Entrepreneurship Program (Hubei Provincial Level)**
Trustworthy Enhancement of LLMs via a Neuro-Symbolic Framework
Project Lead | CNY 20,000 (approx. US\$2,980)

Honors

- **Outstanding Student, Qiming College** (*Highest Honor for HUST Undergraduates; Top 1.85%*) 2025 – 2027
- **Jin Yugang Lan-Ying Scholarship** (*Top 1%*) 2025 – 2026
- **National Scholarship** (*Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide*) 2025 – 2026
- **Triple-A ("Sanhao") Student Scholarship** (*Top 5.5%*) 2025 – 2026
- **National Scholarship** (*Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide*) 2024 – 2025
- **Triple-A ("Sanhao") Student Scholarship** (*Top 5.5%*) 2024 – 2025
- **Outstanding Youth League Member** (*Top 4.1%*) 2024 – 2025
- **Outstanding Youth League Cadre** (*Top 4.1%*) 2024 – 2025
- **Academic Excellence Scholarship & Self-improvement Scholarship** (*Top 10%*) 2024 – 2025

Service

Reviewer, ACL ARR January, May, and August 2026

Academic Activities

ACL 2026 Poster Presenter and Oral Presenter July 1 – 8, 2026
Selected Rising Star Poster Presenter, X-AGI Conference Oct. 17 – 18, 2026 (*upcoming*)

Skills

ML & Research Stack: Python, C/C++, PyTorch, HuggingFace, LoRA, SFT, RLHF, vLLM

Mathematical Foundations: Linear Algebra, Probability Theory, Calculus, Discrete Mathematics

Languages: English (TOEFL 5.5/6; W6 R6 L5.5 S5; CET-4: 609; CET-6: 611), Chinese (Native)

Leadership Experience

- **Class Representative, Qiming Experimental Class 2401** Sept 2024 – Present
 - Organized academic events, managed administrative tasks, and mentored students to enhance class cohesion.
- **Head of Organization Department, School of Cyber Science and Engineering** Sept 2025 – Present
 - Managed administrative and organizational affairs for the Communist Youth League.
- **Head of Lesson Planning, Peer Lecturer Group** Apr 2025 – Present
 - Fostered academic excellence across the college through peer mentoring and curriculum planning.